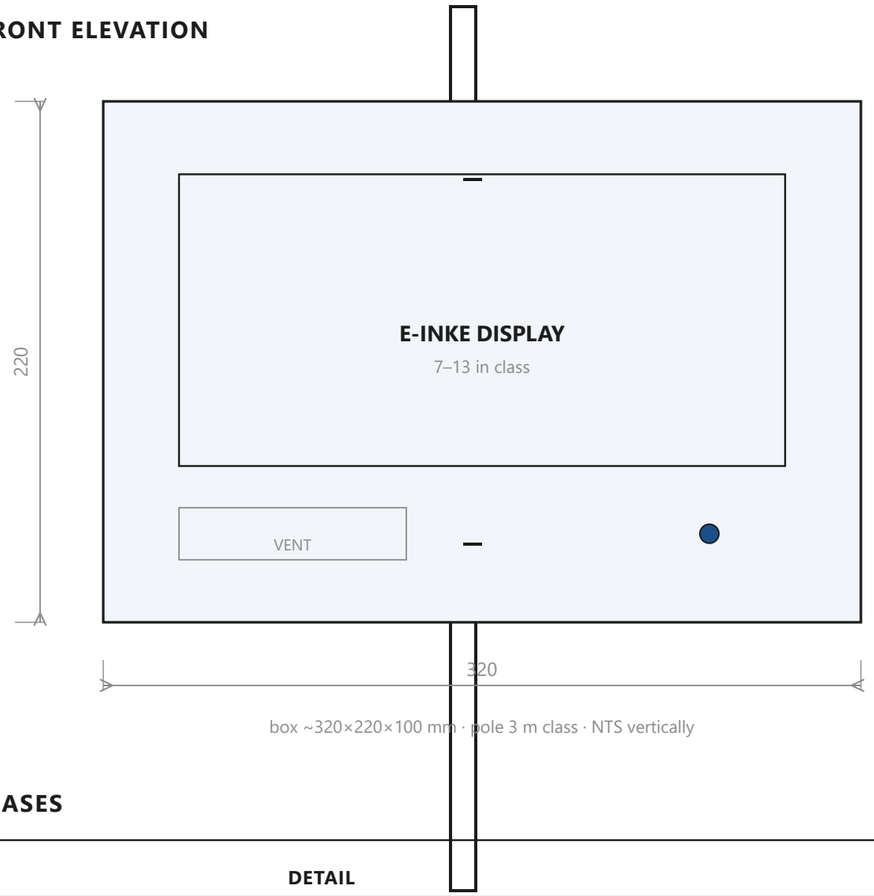
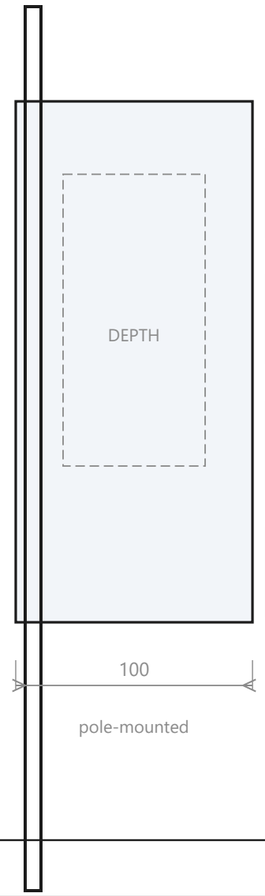
FRONT ELEVATION



box ~320×220×100 mm · pole 3 m class · NTS vertically

SIDE ELEVATION



pole-mounted

CONTENT PIPELINE

PUBLISHER
community / agency signs content

↓

MESH
signed envelopes flood the network

↓

TOTEM
verify → render → E-Ink

↓

PUBLIC
sunlight-readable · zero idle power

USE CASES

USE	DETAIL
Emergency alerts	wildfire evacuation, weather — mesh-wide in minutes
Community notices	council, school, clinic, events — the digital general-store board
Agency bulletins	province, RCMP, health unit — signed content only
Local info	road conditions, water advisories, powwow/community events
Mesh status	optional node-health display — coverage made visible

WHY IT'S UNIQUE

— Only public noticeboard that IS a network node — it routes, repeats, stores.

— Signed content: every notice carries publisher identity — no anonymous flyers.

— Works when grid AND cell network are both down — the blackout case.

— EN/FR pipeline; community controls what appears — sovereignty in public space.

ARCTICCHAIN — RURAL NODE FLEET

EMEBET INC. · emebetos.ca
SOLE PROPRIETARY OWNER — SURAFEL NEGUSSIE, PLANNER/MAKER

DRAWN BY **Surafel Negussie**
CHECKED BY
DATE
PROJECT

SCALE **NTS**
SHEET **1/2**
DWG NO **AF-E-101**
UNITS mm

PROPRIETARY & CONFIDENTIAL — UNAUTHORIZED DISTRIBUTION PROHIBITED

PARTS LIST — PER TOTEM

#	PART	SUPPLIER	APPROX CAD	QTY
1	CM5 4GB + IO Board (same as L1 — fleet common)	PiShop / Canada Computers	$90–130	1
2	13.3in or 7.5in e-Paper HAT (SPI)	Waveshare / DFRobot	$100–200	1
3	SX1262 LoRa breakout 915 MHz	DFRobot / AliExpress	$15–25	1
4	SIM7600 LTE HAT (optional backhaul)	DFrobot / Waveshare	$60–90	0–1
5	20–30 W poly solar panel + MPPT 20 A	Amazon.ca / Renogy	$120–200	1
6	12.8 V 20 Ah LiFePO4 battery	Amazon.ca / supplier	$180–260	1
7	IP66 polycarbonate box ~320×220×100	Bud Industries / Amazon.ca	$80–140	1
8	Pole + U-bolts + window film + gasket	hardware supplier	$60–100	1

POWER SYSTEM

20–30 W PANEL → south tilt
MPPT 20 A → LiFePO4 profile
LiFePO4 20 Ah → ≈256 Wh
5 V BMS → CM5 → always-on

Autonomy ≈3–5 days no-sun at 2–4 W avg; oversized panel is the winter plan.

INSTALLATION — TWO PERSON, ~30 MIN

1. Set pole (post-hole or existing utility pole U-bolt).
2. Hang box on rear hinge; face display to road/foot traffic.
3. Connect solar leads; route antenna ports top, drip-loop cable.
4. Power on; verify mesh join + first content render.
5. Service: front latch → SD card + battery swap <10 min.

CONTENT PIPELINE — SIGNED PUBLICATIONS

— Publishers sign content (Ed25519) — community/agency keys.
— Mesh floods signed notice envelopes; node verifies before render.
— Rotation: alerts > notices > info; expiry timestamps auto-clear.
— Rate-limit + dedup; offline queue preserves order; admin can pin/hide.

ARCTICCHAIN — RURAL NODE FLEET

EMEBET INC. · emebetos.ca
SOLE PROPRIETARY OWNER — SURAFEL NEGUSSIE, PLANNER/MAKER

DRAWN BY **Surafel Negussie**
CHECKED BY
DATE
PROJECT

SCALE **NTS**
SHEET **2/2**
DWG NO **AF-E-102**
UNITS mm

PROPRIETARY & CONFIDENTIAL — UNAUTHORIZED DISTRIBUTION PROHIBITED